



Reactor

Monitor and Optimize Your Solutions

**Get Certified: DP-700 Data Engineer
from Synapse to Fabric (PST/GMT)**



Just Blindbæk
Microsoft Data Platform MVP



Thomas Martens
Microsoft Data Platform MVP



Reactor

Just Blindbæk

Microsoft Data Platform MVP

JustB.dk/blog

linkedin.com/in/blindbaek/





Reactor

Thomas Martens

Microsoft Data Platform MVP

<https://www.linkedin.com/in/tommartens68/>

<https://publish.obsidian.md/minceddata/Minceddata/About+me>





Reactor

Meet your moderators

(Live Only)

They are in the chat available to answer your questions



Markus Ehrenmueller-Jensen
Microsoft Data Platform MVP



Ginger Grant
Microsoft Data Platform MVP



Reactor

Get Certified: From Synapse to Fabric

Date	Title
Tuesday January 28	Implement and manage analytics solutions
Wednesday January 29	Ingest and transform data
Thursday January 30	Monitor and optimize your solutions
Wednesday February 5	Prepare for exam day and open Q&A

aka.ms/dp700/s1a

Exam DP-700
Fabric Data Engineer



MGM GRAND, Las Vegas, NV

**March 31 -
April 2, 2025**

Workshops: March 29, 30, and April 3

Join us at the second annual Microsoft Fabric Community Conference and get up close with the latest data, analytics, and AI developments—plus network with community leaders and other technical experts from around the world.

Use code MSCUST for a \$150 discount. Prices go up February 11th.



Register now: aka.ms/fabcon25/dp700

50% discount for Exam DP-700

Ends March 31, 2025

Limited time offer, while supplies last

aka.ms/gc/dp700

Get certified in Microsoft Fabric—for half-price

Limited-time offer!

Accelerate your success with a 50% discount Microsoft certification exam voucher

Get ready to fast-track your career in data and AI by earning your Microsoft Certified: Fabric Data Engineer Associate certification (Exam DP-700). There's no faster way to climb the ladder than with industry-recognized Microsoft certifications.

To be eligible for this [limited-time](#) offer, you must:

- [Join](#) the Fabric Community if you haven't already.
- Have a [Microsoft Learn profile](#) and be able to provide your Microsoft Learn [username](#).
- Not already be a Microsoft Certified: Fabric Data Engineer Associate (Exam DP-700).
- Have achieved one of the following:
 1. Previously passed exam DP-203, [or](#)
 2. Completed all modules in this [Fabric learning plan](#), [or](#)
 3. Attended one of the [Get Certified: DP-700 live or on demand learning sessions](#).
- Be confident that you can schedule and pass exam DP-700 within 30 days of receiving your 50% discount voucher.
- Agree to these [terms and conditions](#).

Submit your request form as soon as possible. Exam discount voucher quantities are limited, so don't wait.

Only one request per person will be accepted. If you are eligible, you will receive your voucher via private message in the Fabric Community 7-10 days after submitting the request form. Your voucher will expire 30 days after we send you the voucher. You must register for and complete the exam before the expiration date.

The window for submitting your request form ends on March 31, 2025, and we will not accept any request forms after that date.

For additional details, check out the frequently asked questions below. Your participation in this offer is governed by these [terms and conditions](#).

Frequently asked questions

- > How do I create a Fabric Community account?
- > Where do I find my private messages on the Fabric Community?
- > What is the deadline for submitting the request form by?
- > How will I know if I'm eligible or not eligible?

Yes, I am ready to take the DP-700 exam!

Fabric Community username

FabricPam

Country/location *

Select

Microsoft Learn [username](#) *

Username not display name

50 characters max

Why do you want to earn a Fabric Data Engineer Associate certification? *

Provide your answer without using special characters such as %, \$, and #.

300 characters max

I have prepared for the exam by doing atleast one of the following: *

- ☐ Completing all modules in this [Fabric learning plan](#).
- ☒ Attending one of the [Get Certified: DP-700 live or on demand learning sessions](#).
- ☐ Previously passing exam DP-203.

I affirm that I: *

- ☐ Have not already passed exam DP-700.
- ☐ Am confident that I can take and pass exam DP-700 within 30 days of receiving the 50% discount voucher.
- ☐ Agree to these [terms and conditions](#).

Request my 50% discount exam voucher

Live & interactive

Say "hi" and ask questions in the chat



Get Certified: DP-700



Monitor and Optimize Microsoft Fabric Solutions

aka.ms/dp700studyguide

What we'll cover today

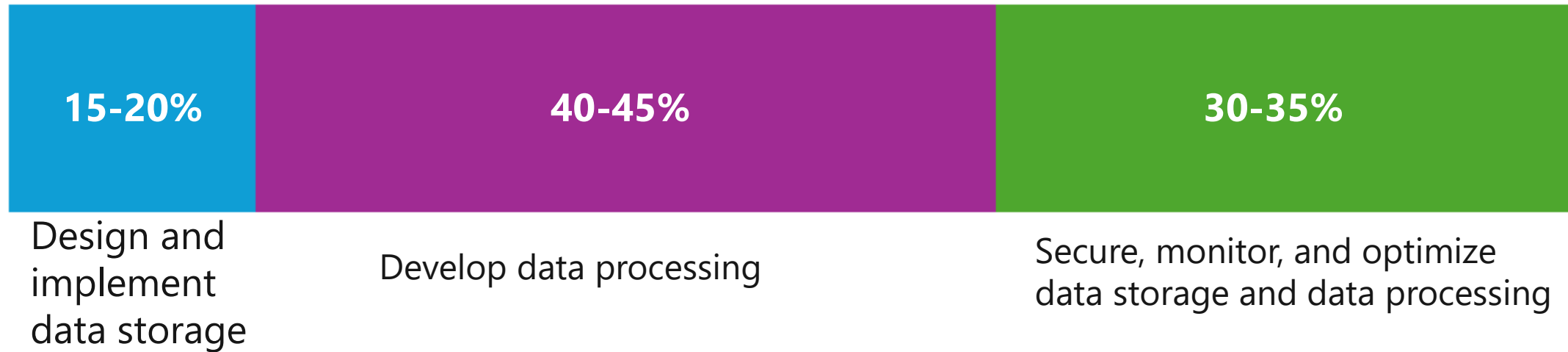


- DP-203 and DP-700 skill comparison
- Intro to Microsoft Fabric
- Why monitoring and optimization matter in Fabric
- Monitor analytics solutions in Fabric
- Optimize analytics solutions in Fabric
- Security and compliance monitoring
- Practical tips for success in DP-700

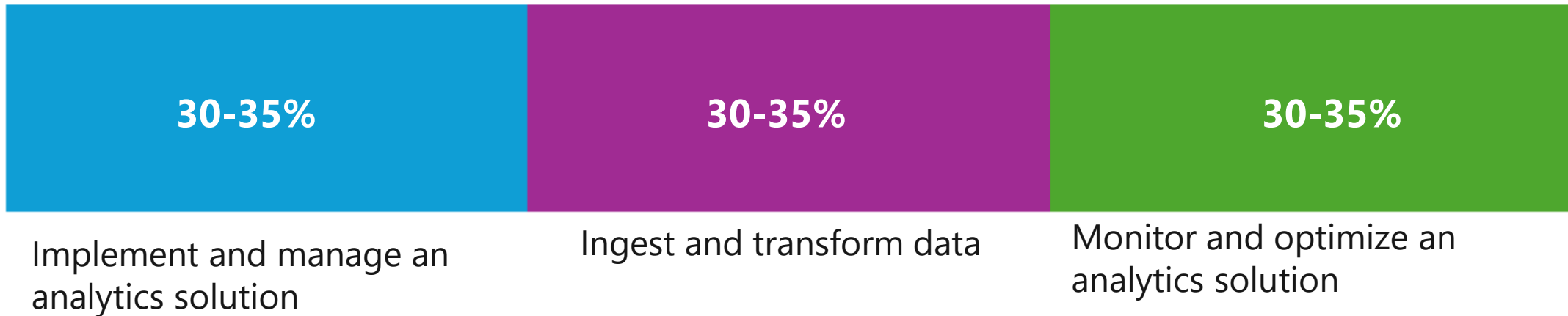
DP-203 and DP-700 skill comparison



DP-203: Data Engineering on Microsoft Azure



DP-700: Data Engineering using Microsoft Fabric



DP-203: Data Engineering on Microsoft Azure

Design and implement data storage (15–20%)

- Implement a partition strategy
- Design and implement the data exploration layer

Develop data processing (40–45%)

- Ingest and transform data
- Develop a batch processing solution
- Develop a stream processing solution
- Manage batches and pipelines

Secure, monitor, and optimize data storage and data processing (30–35%)

- Implement data security
- Monitor data storage and data processing
- Optimize and troubleshoot data storage and data processing

DP-700: Data Engineering using Microsoft Fabric

Implement & manage an analytics solution (30–35%)

- Configure Microsoft Fabric workspace settings
- Implement lifecycle management in Fabric
- Configure security and governance
- Orchestrate processes

Ingest and transform data (30–35%)

- Design and implement loading patterns
- Ingest and transform batch data
- Ingest and transform streaming data

Monitor and optimize an analytics solution (30–35%)

- Monitor Fabric items
- Identify and resolve errors
- Optimize performance

Key differences between DP-203 and DP-700

	DP-203 (Azure Data Engineer)	DP-700 (Fabric Data Engineer)
Scope of Services	Broad range of Azure services (e.g., Synapse, Data Lake, Stream Analytics)	Focused on Microsoft Fabric, a unified analytics platform (SaaS)
Primary Focus	Data storage and processing pipelines	Building and optimizing analytics solutions on Fabric
Tooling	Azure-specific tools like Data Factory, Databricks, Event Hubs	Fabric-native tools like OneLake, Lakehouse, Data Warehouse, Eventhouse
Target Audience	Engineers working across diverse Azure ecosystems	Engineers transitioning to or working in Fabric
Experience	Navigates multiple services and interfaces	Streamlined, unified platform managed in Fabric portal

Introduction to Microsoft Fabric



Unified data platform for the era of AI



Data Factory



Data Engineering



Data Warehouse



Data Science



Real-Time Intelligence



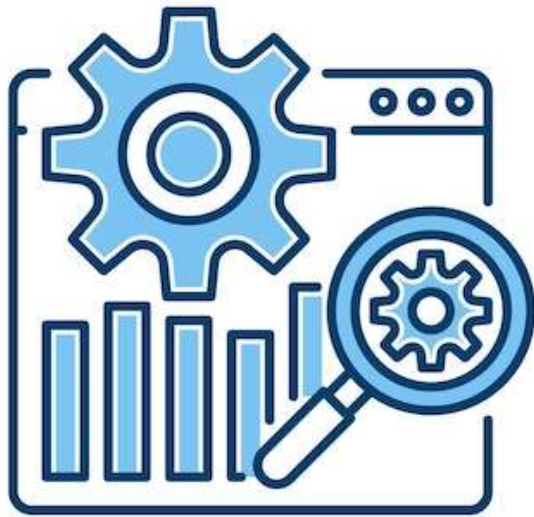
Power BI

- **Data Factory:** Data integration combining Power Query with the scale of Azure Data Factory to move and transform data.
- **Data Engineering:** Data engineering with a Spark platform for data transformation at scale.
- **Data Warehouse:** Data warehousing with industry-leading SQL performance and scale to support data use.
- **Data Science:** Data science with Azure Machine Learning and Spark for model training and execution tracking in a scalable environment.
- **Real-Time Intelligence:** Real-time Intelligence to query and analyze large volumes of data in real-time.
- **Power BI:** Business intelligence for translating data to decisions.

Why monitoring and optimization matter in Fabric



Significance of monitoring and optimization in Fabric



- Maximize performance and efficiency
- Cost control
- Better resource allocation
- Enhanced data throughput and timeliness
- Proactive issue detection and resolution

Benefits of monitoring and optimization

- Scale more efficiently
- Reduce risk of data corruption or loss
- Improve user experience
- Maximize your capacity
- Set up automated alerts



Monitor analytics solutions



Monitor Fabric solutions

Data pipelines

Dataflows

Semantic
model
refreshes

Spark jobs

Notebooks

Lakehouses

Eventstreams

Monitoring best practices



Identify what
to monitor



Collect and
review data



Resolve
issues



Optimize
performance

Fabric monitoring hub

View details

- Activity status
- Start and end time
- Duration
- Activity statistics

Monitor

View and track the status of the activities across all the workspaces for which you have permissions within Microsoft Fabric.

Refresh

Filter by keyword

Filter

Column Options

Clear all

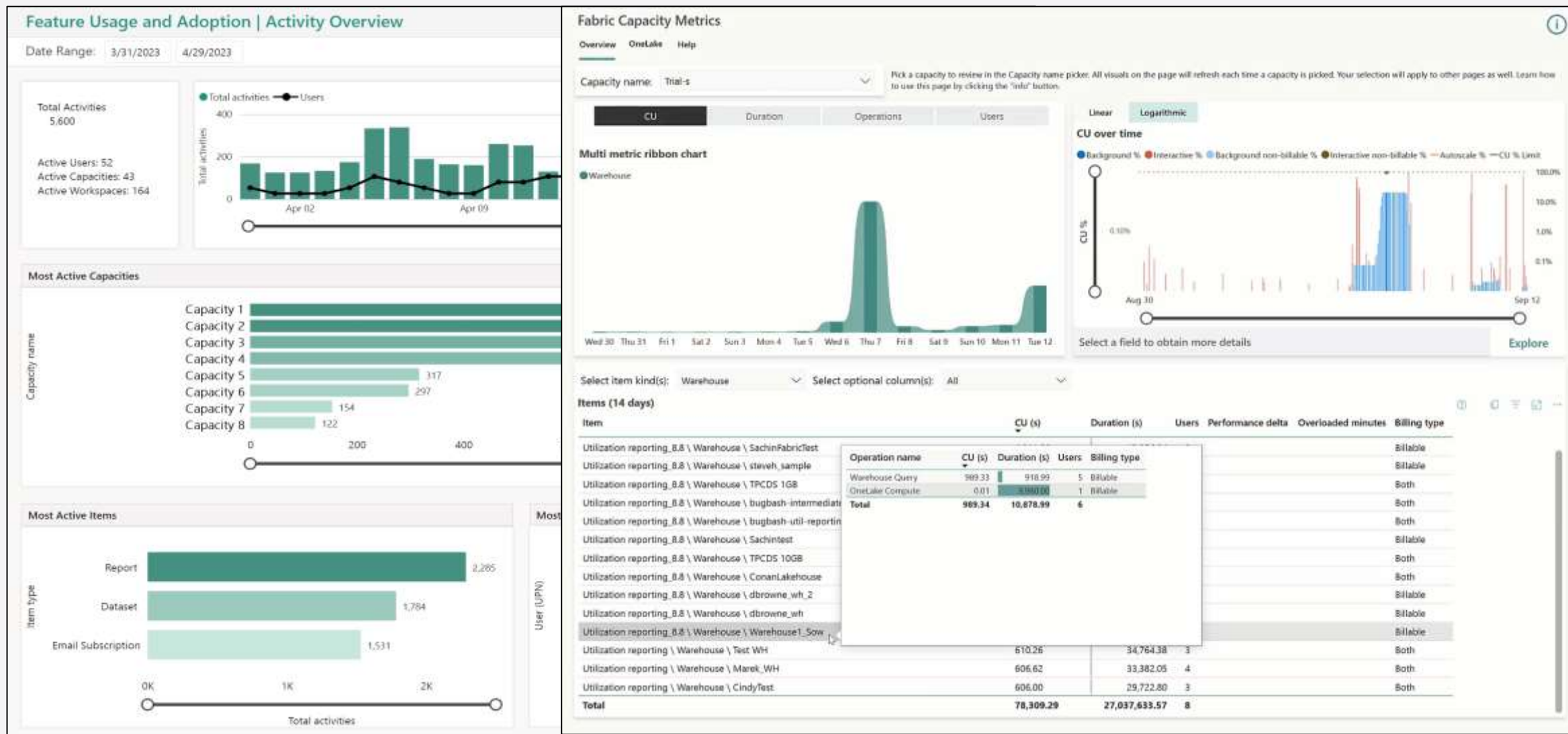
To apply filters, select the values from the Filter dropdown menu.

Activity name	Status	Item type	Start time	Submitted by	Location
Purview Hub	Succeeded	Semantic model	12/12/2024, 2:44 PM	Admin Monitoring	Adm
Feature Usage and Adoption	Succeeded	Semantic model	12/12/2024, 2:44 PM	Admin Monitoring	Adm
LHDP0000	Succeeded	Semantic model	12/11/2024, 11:59 AM	System Administrator	fab_v
semantic model refresh	Succeeded	Data pipeline	12/11/2024, 11:58 AM	System Administrator	fab_v
GetProductData	Succeeded	Dataflow Gen2	12/09/2024, 4:08 PM	System Administrator	fab_v
Query products_305df121-cd8f-49a2-8c7a-2203...	Succeeded	Notebook	12/09/2024, 4:03 PM	System Administrator	fab_v
pipeline3	Succeeded	Data pipeline	12/05/2024, 5:20 PM	System Administrator	Dem

Additional monitoring capabilities for admins

Monitor your environment

Monitor capacity usage



Optimize analytics solutions



Optimization in Fabric

- Key focus areas:
 - storage (lakehouses, data warehouses, databases)
 - compute (Spark)
 - processing (eventstreams and pipelines)
- Fabric connects everything through OneLake and a unified compute engine, so optimization techniques can have a cascading effect on overall performance and costs

Storage optimization

- Lakehouse Tables:
 - Use **Delta Lake optimization** techniques: optimize command, compaction, Z-order clustering for faster queries
 - Implement **partitioning and caching** to enhance read performance for large datasets
- Data Warehouses:
 - Analyze and adjust table storage formats
 - Use **columnstore indexes** for efficient analytical queries
 - Monitor distribution keys and partition strategies to balance performance across nodes

Compute and processing optimization

- **Pipelines**

- Optimize data movement by limiting unnecessary transformations and leveraging parallelism.
- Use incremental loads instead of full loads.
- Configure pipeline triggers efficiently (event-based triggers for real-time scenarios).

- **Real-time data - Eventstreams & Eventhouses**

- Reduce data volume with real-time filtering and transformations at ingestion (e.g., apply filters and aggregations).
- Use time windows to group streaming data efficiently.

- **Spark**

- Optimize cluster sizing based on workload needs (e.g., autoscale for cost control).
- Tune partition sizes, caching, and broadcast joins for query optimization.

SQL Query optimization

- **Leverage Materialized Views**

- Use materialized views for datasets that are frequently queried to improve performance.

- **Indexing and Query Hints**

- Apply indexing and query hints where applicable to speed up query execution.

- **Optimize Query Patterns**

- Reduce joins and push down filters to streamline queries.

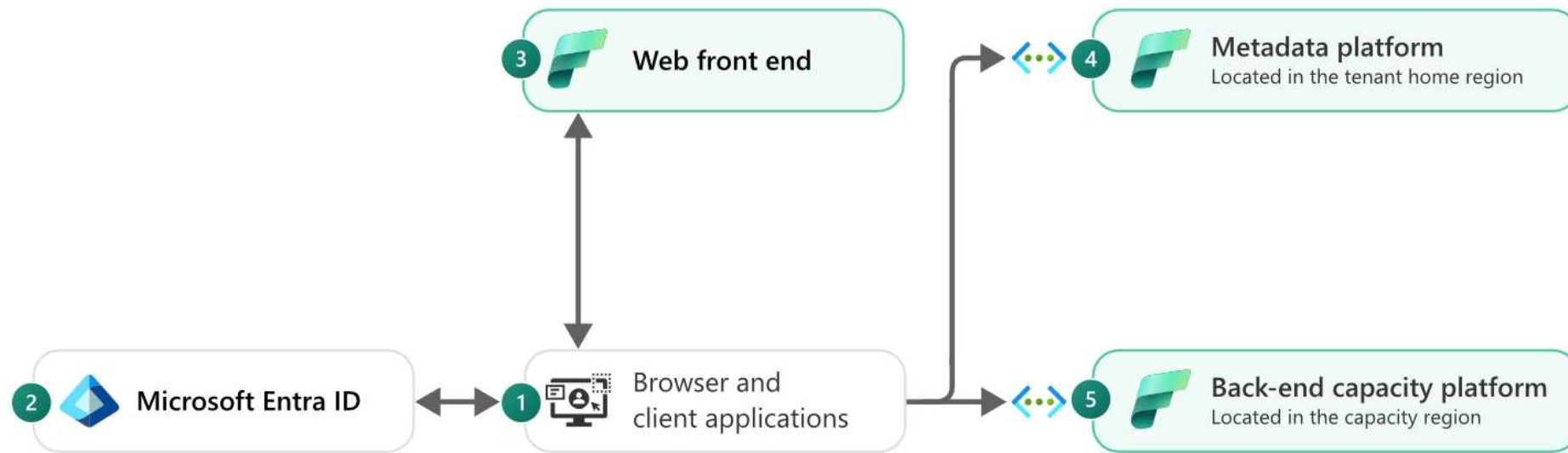
- **Monitor Query Plans**

- Utilize Fabric's built-in tools to monitor query plans and execution, gaining insights for further optimization.

Security and compliance monitoring



Security in Microsoft Fabric



Data security in Fabric

- **Fabric Integration:** Operates within your Microsoft Entra tenant, ensuring logical data isolation across regions and secure multi-tenant management within the Microsoft ecosystem.
- **Encryption at Rest:** Data stored in OneLake is encrypted, ensuring security while at rest.
- **Data in Transit:** All data transferred within Fabric is encrypted using TLS for secure communication.
- **Role-Based Access Control (RBAC):** Granular permissions restrict access to data and resources.

Data compliance in Fabric

- **Regional Data Storage:**
 - Data at rest is stored in your home region or in remote regions of your choice.
 - For multi-geo tenants, data is stored within the geo where it was created to meet compliance needs.
- **Metadata and Processing:** Securely managed in the tenant's home geo.
- **Compliance Certifications:** Meets major industry standards (e.g., GDPR, HIPAA, SOC).
- **Auditing and Monitoring:** Tracks access and changes to enhance governance.
- **Private Endpoints:** Secure network-level isolation through Azure Private Link.

Monitoring security and compliance

- Activity logs
- Audit logs
- Data access monitoring
- Alerts and notifications
- Integration with monitoring tools
- Compliance reports

Practical tips for success in DP-700



DP-700 Exam: Monitor and optimize Fabric solutions

- Hands-on practice
- Study guides and resources
- Practice tests on learn
- Microsoft Learn course
- Sample projects

aka.ms/fabrictrial



Reactor

Get Certified: From Synapse to Fabric

Date	Title
Tuesday January 28	Implement and manage analytics solutions
Wednesday January 29	Ingest and transform data
Thursday January 30	Monitor and optimize your solutions
Wednesday February 5	Prepare for exam day and open Q&A

aka.ms/dp700/s1a

Exam DP-700
Fabric Data Engineer



MGM GRAND, Las Vegas, NV

**March 31 -
April 2, 2025**

Workshops: March 29, 30, and April 3

Join us at the second annual Microsoft Fabric Community Conference and get up close with the latest data, analytics, and AI developments—plus network with community leaders and other technical experts from around the world.

Use code MSCUST for a \$150 discount. Prices go up February 11th.



Register now: aka.ms/fabcon25/dp700

50% discount for Exam DP-700

Ends March 31, 2025

Limited time offer, while supplies last

aka.ms/gc/dp700

Get certified in Microsoft Fabric—for half-price

Limited-time offer!

Accelerate your success with a 50% discount Microsoft certification exam voucher

Get ready to fast-track your career in data and AI by earning your Microsoft Certified: Fabric Data Engineer Associate certification (Exam DP-700). There's no faster way to climb the ladder than with industry-recognized Microsoft certifications.

To be eligible for this [limited-time](#) offer, you must:

- [Join](#) the Fabric Community if you haven't already.
- Have a [Microsoft Learn profile](#) and be able to provide your Microsoft Learn [username](#).
- Not already be a Microsoft Certified: Fabric Data Engineer Associate (Exam DP-700).
- Have achieved one of the following:
 1. Previously passed exam DP-203, [or](#)
 2. Completed all modules in this [Fabric learning plan](#), [or](#)
 3. Attended one of the [Get Certified: DP-700 live or on demand learning sessions](#).
- Be confident that you can schedule and pass exam DP-700 within 30 days of receiving your 50% discount voucher.
- Agree to these [terms and conditions](#).

Submit your request form as soon as possible. Exam discount voucher quantities are limited, so don't wait.

Only one request per person will be accepted. If you are eligible, you will receive your voucher via private message in the Fabric Community 7-10 days after submitting the request form. Your voucher will expire 30 days after we send you the voucher. You must register for and complete the exam before the expiration date.

The window for submitting your request form ends on March 31, 2025, and we will not accept any request forms after that date.

For additional details, check out the frequently asked questions below. Your participation in this offer is governed by these [terms and conditions](#).

Frequently asked questions

- > How do I create a Fabric Community account?
- > Where do I find my private messages on the Fabric Community?
- > What is the deadline for submitting the request form by?
- > How will I know if I'm eligible or not eligible?

Yes, I am ready to take the DP-700 exam!

Fabric Community username

FabricPam

Country/location *

Select

Microsoft Learn [username](#) *

Username not display name

50 characters max

Why do you want to earn a Fabric Data Engineer Associate certification? *

Provide your answer without using special characters such as %, \$, and #.

300 characters max

I have prepared for the exam by doing atleast one of the following: *

- ☐ Completing all modules in this [Fabric learning plan](#).
- ☒ Attending one of the [Get Certified: DP-700 live or on demand learning sessions](#).
- ☐ Previously passing exam DP-203.

I affirm that I: *

- ☐ Have not already passed exam DP-700.
- ☐ Am confident that I can take and pass exam DP-700 within 30 days of receiving the 50% discount voucher.
- ☐ Agree to these [terms and conditions](#).

Request my 50% discount exam voucher

Questions?

Find resources at aka.ms/iamready/dp700prepare
Continue the conversation at aka.ms/forum/dp700

